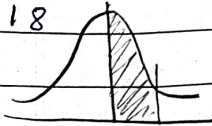


Assignment 3

1. $n = 50$, $\bar{x} = 2550$, $\mu = 2500$, $\sigma = 300$

$$Z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} = \frac{2550 - 2500}{300/\sqrt{50}} = 1.18$$



from Z-score table, Z-score = 0.881

$$P(Z > 1.18) = 1 - 0.881 = 0.119$$

2. $n = 100$, $\sigma^2 = \frac{1}{4}$, $\mu = 2$

$$P(190 < X_1 + X_2 + \dots + X_{100} < 210)$$

Let $X_1 + X_2 + \dots + X_{100} = X$

$$P(190 < X < 210) = ?$$

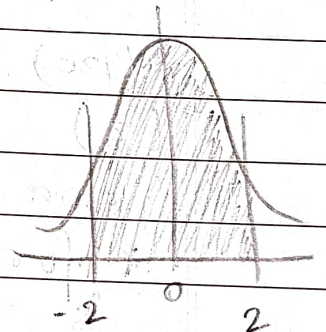
Mean of $X = 100 \times 2 = 200$

Variance of $X = 100 \times \frac{1}{4} = 25$

Standard deviation of $X = \sqrt{25} = 5$

for $X = 190 \rightarrow Z = \frac{190 - 200}{5} = -2$

$X = 210 \rightarrow Z = \frac{210 - 200}{5} = 2$



$$P(-2 < Z < 2) = 0.9772 - 0.0228 = 0.9544$$

NAME: _____

STD.: _____

DIV.: _____

3. $n = 144$, $\mu = 30 = \sigma$ $\left(\mu = \frac{1}{\lambda}, \quad V = \frac{1}{\lambda^2} \rightarrow \mu = \sigma \right)$

Total hrs = 4500

Mean ($\bar{x}$) = $\frac{4500}{144} = 31.25$

$\mu = 144 \times 30 = 432$

$\sigma = 12 \times 30 = 360$

$z = \frac{31.25 - 30}{30/\sqrt{144}} = 0.5$

$z = \frac{\bar{x} - 4320}{360} = \frac{4500 - 4320}{360}$

$P(z > 0.5) = 1 - 0.6915 = \boxed{0.3085}$

4. $\lambda = 2 = \mu = \sigma^2$, $n = 75$

$P(120 < S_n < 160)$

$z = \frac{\bar{x} - \mu}{s/\sqrt{n}} \rightarrow \frac{\bar{x} - 150}{\sqrt{2 \times 75}} = \frac{\bar{x} - 150}{\sqrt{150}}$

$P(120 < S_n < 160) = ?$

$P(160) = \frac{160 - 150}{\sqrt{150}} = 0.816 \approx 0.82 \rightarrow z = 0.7939$

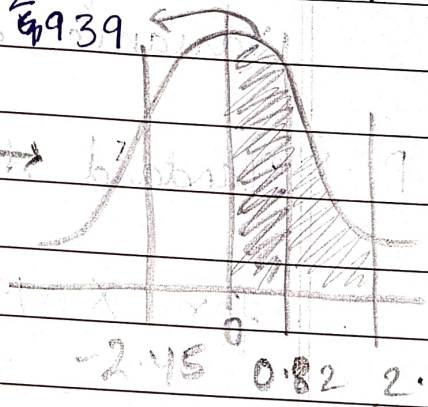
$P(120) = \frac{120 - 150}{\sqrt{150}} = -2.45$

score

z for 2.45 = 0.9927

z -score for 2.45 = 0.9927 - 0.5 = 0.4927

total z score = 0.4927 + 0.2939 = $\boxed{0.7868}$



5. $n = 50, \mu = 2, \sigma^2 = 1$

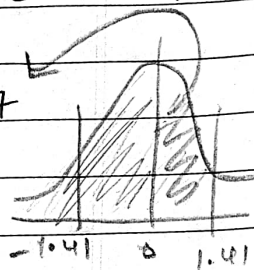
Total mean = $50 \times 2 = 100$

Total variance = $50 \times 1 = 50 \rightarrow$ standard dev sum = $\sqrt{50}$

$P(90 < Y < 110)$

$P(110) \rightarrow z \text{ for } 110 = \frac{110 - 100}{\sqrt{50}} = 1.41 \rightarrow z\text{-score} = 0.9207$

$P(0 < z < 1.41) = 0.9207 - 0.5 = 0.4207$



~~$z \text{ for } 90 = \frac{90 - 100}{\sqrt{50}} = -1.41 \rightarrow z\text{-score}$~~

$P(-1.41 < z < 1.41) = 0.4207 \times 2 = \boxed{0.8414}$

6. $n = 100, \mu = 29321, \sigma = 2120$

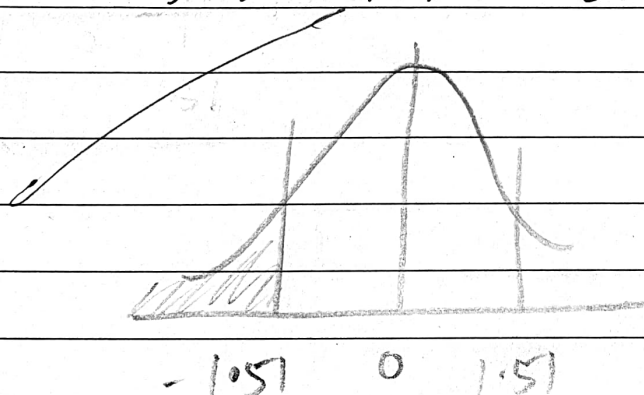
$P(\bar{x} < 29000) = \frac{29000 - 29321}{2120/\sqrt{100}} = -1.51$

$P(z < 1.51) = 0.9345$

$P(-1.51 < z < 1.51) = (0.9345 - 0.5) \times 2 = 0.869$

$P(z < -1.51) = 0.9345 - 0.869 = 0.0655$

$= \boxed{6.55\%}$



RNP
22/01